

ZHACHORY VOLKER

hello@zhach.me | (940) 315-4580

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Staff Software Engineer with 10+ years of professional full-stack and Machine Learning experience. Proven track record of designing scalable Search and Recommendation stacks, optimizing robust data infrastructure, and drastically reducing compute overhead. Specialized in building high-impact ML ranking architectures at massive scale and leveraging modern AI development tools to accelerate delivery and business growth.

Connect

Github: @Zhachory1

LinkedIn: zhachory1

Website: zhach.me

Experience

Staff Software Engineer *Rokt*

October 2025 - Present | New York, New York

- Reduced machine learning audience generation pipeline costs by 75% (saving ~\$750k annually) by migrating to a highly stable, efficient two-tower infrastructure and implementing a robust data refresh system.
- Architected and deployed a dynamic ranking layer for on-the-fly audience generation, optimizing for CVR and scaling revenue from generated audiences by 120%.
- Led technical architecture and hands-on engineering for the Audience and Bidding domain, delivering a 40% uplift in CVR for ML-generated audiences by building independent, scalable solutions outside of legacy dependencies.
- Spearheaded an org-wide AI integration initiative, upskilling engineers to utilize AI tooling (Claude, Cursor, OpenCode, Codex) for coding, documentation, system management, and debugging, improving overall Developer Satisfaction by 25%.
- Founded and led the Deaf affinity group, championing "Deaf Gain" to leverage unique visual skills and promote workplace diversity and inclusion.

Senior Software Engineer *YouTube Search*

March 2018 - September 2025 | New York, New York

- Led a resource optimization project achieving a 60% reduction in compute usage through efficient metric analysis and ranking model refactoring (estimated annual savings: \$1.8M).
- Designed and implemented a unified ranking framework to evaluate diverse content types (shelves, playlists, channels, articles) within the same scoring space on the YouTube Search results page, increasing overall Search Active Users (SAUs) by 2% and increasing CTR by an average of 25% for multiple features in a single launch.
- Created the YouTube Search News experience by implementing clustering algorithms, NLP models, and scalable infrastructure, resulting in a 38% increase in News Click-Through Rate (CTR) and a 10% rise (~8 Million queries) in News Searches per day.
- Launched a critical feature surfacing authoritative news articles for breaking news queries to combat misinformation, collaborating with Google News and improving news user satisfaction by 13%.
- Collaborated cross-functionally with teams across YouTube and Google to improve code health, testing practices, and developer tooling, reducing code complexity (Halstead score) by 9% and increasing internal documentation health score by 25%.

Software Engineering Resident *Google Research*

March 2017 - March 2018 | New York, New York

- Applied NLP and Machine Learning to 10,000+ articles (RNNs, CNNs, Decision Trees) for associating user questions with relevant documents in the fact-checking corpus, which can be seen on the first page of Google current day.
- Created and maintained MapReduce pipelines (C++) for managing the size and processing of the fact-checking article corpus which was used by 6 Large News Publishers with XXM sized audiences.
- Contributed to the maintenance and enhancement of a graph mining library utilized across Ads, YouTube, Research, and Maps.
- Developed data pipelines using label propagation and semi-supervised algorithms on large graphs, increasing library adoption by 8%.
- Developed an internal research tool using Java, TypeScript, and Angular to facilitate discovery of fact-checking articles still used by 100s of journalists today.
- Implemented robust evaluation functions for effective model training based on a published research paper's ("Watch Your Step: Learning Node Embeddings via Graph Attention") internal implementation.

Education

University of North
Texas

Bachelors of Computer
Science

August 2013 - May 2018

GPA: 3.443

Relevant Coursework:

Advanced Algorithms, Data
Structures, Software
Engineering, Discrete Math,
Game Mechanics, Operating
Systems, Artificial
Intelligence

Dean's List:

Fall 2013, Spring 2015, Fall
2015, Spring 2016

Skills

Machine Learning & AI: TensorFlow, PyTorch, Deep
Learning, NLP, Transformers, Reinforcement Learning,
Genetic Algorithms, Clustering, Label Propagation, Semi-
Supervised Learning, Graph Mining, Embeddings,
Ranking Systems, Fine-Tuning, RAG, Evals

Agentic AI: Claude Code, Codex, Cursor, OpenCode,
Coding Agents, Prompt Engineering, Loop & Context
Engineering, MCP Servers & Tooling, Subagents, Skills &
Workflows, Agent Evals

Software Development & Infrastructure: Scalable
Infrastructure Design, System Monitoring, CI/CD, Docker,
Kubeflow, Buildkite, Object-Oriented Programming
(OOP), Test-Driven Development (TDD), Advanced
Algorithms, Data Structures, Code Health Optimization

Soft Skills: Leadership & Technical Mentorship,
Collaboration & Teamwork, Communication, Problem-
Solving, Conflict Resolution, Analytical Thinking,
Adaptability, Continuous Learning

Open Source

Me Write Code — Token-efficient
terminal coding agent. 20+
providers, plan mode, subagents,
MCP, and hooks.

Autopraxis — Self-improving
workflow skills for AI agents, turning
messy goals into grounded briefs,
reviewed plans, shipped work, and
measured learning loops.

agent-fleet — A portable council of
specialist review personas for high-
stakes engineering decisions.

ZBrain — Local-first markdown / doc
RAG for private work context.

kns — Hierarchical RAG with DNS-
like resolution across private, team,
and company knowledge zones.